Circularity Data Exchange Landscape analysis and way forward

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Foreword

In today's rapidly evolving business and regulatory landscape, the transition to a circular economy has become an imperative for sustainable growth and resilience. Achieving this vision requires the ability to exchange consistent, credible data across value chains. However, fragmented approaches to defining, operationalizing, and sharing circularity data currently hinder progress.

This White Paper consolidates insights from The Circularity Data Exchange's (CDX) Scoping Phase - an exploration into the World Business Council for Sustainable Development's (WBCSD) role in facilitating the exchange of credible, consistent, and comparable circularity data, at the product level. Launched in January 2024 and with support from KPMG and Circularise, the Scoping Phase sought to explore practical solutions for overcoming data challenges. Through expert interviews, regulatory reviews, and case studies, the report provides insights into the potential for harmonized data exchange in the circular economy. One notable aspect of the report is the consideration of how WBCSD's Partnership for Carbon Transparency (PACT)—a proven methodology for exchanging Scope 3 carbon emissions data— can inform similar efforts for circularity data.

This report does not seek to identify gaps, but serves a foundational resource to guide the next steps toward harmonized circularity data approaches. As such, the results of this white paper serve to inform the direction of the CDX work for 2025 and beyond.

Aligned with WBCSD's broader Circular Economy portfolio, including initiatives like the Circular Transition Indicators framework and the Global Circularity Protocol for Business, CDX offers the data exchange infrastructure needed to operationalize and scale these frameworks. Together, these initiatives lay the groundwork for accelerating the global shift toward circular business models.

To promote the harmonization of circularity data at a global level, the results of the Scoping Phase contribute directly to international alignment efforts. Specifically, the CDX contributes to the Digitalization 4 Circular Economy (D4CE) initiative, co-hosted by One Planet Network and the Coalition for Digital Environment Sustainability (CODES). Under the initiative, CDX supports the development of a workpiece on Global Digital Standards, which focuses on improving the transfer of and access to data along the value chain to inform and accelerate circular economy approaches.

This White Paper highlights the transformative potential of harmonized circularity data exchange, underscoring the importance of global collaboration in building a more resilient, transparent, and sustainable economy.

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Executive Summary

The circular economy is an opportunity to achieve social, economic and environmental sustainability goals. It directly addresses the interconnected crises of climate change, nature and biodiversity loss, and pollution and waste, offering a pathway to mitigate these global challenges. By redesigning products, reducing resource extraction and minimizing waste, the circular economy contributes to achieving net-zero goals and supports sustainable development through economic innovation and green job creation.

With global policies increasingly prioritizing circularity, there is a mounting regulatory push for businesses to accurately report circular practices and product-level data. Thus, WBCSD has explored its role in enabling the exchange of credible, consistent and comparable circularity data at the product level through the Scoping Phase of its Circularity Data Exchange (CDX). This effort offers a glimpse into how comprehensively regulatory requirements and frameworks support the realization of the circular economy's potential and what it means for companies in terms of the data they need to collect to meet them.

WBCSD's review of key regulations and frameworks, such as the European Union's Ecodesign for Sustainable Products Regulation (ESPR) and the International Organization for Standardization (ISO) Product Circularity Data Sheet (PCDS), emphasizes the need for high-quality, standardized circularity data. However, the analysis has identified critical challenges, including fragmented definitions, inconsistent metrics and complex requirements across frameworks and across value chains. This variation in circularity data standards complicates efforts to manage product-level data consistently and undermines cross-sector collaboration, making it challenging for companies to comply with and compare circularity efforts.

In addition to methodology challenges, an effective circular economy demands a robust data exchange system – one that allows the seamless sharing of circularity data across global supply chains and industries. Recognizing the need for companies to measure their progress towards a circular economy, WBCSD has developed the Circular Transition Indicators (CTI) – a universal framework of metrics to measure circular performance along a company's value chain and understand how to reduce impacts on climate and nature.

Building on the CTI, WBCSD and the One Planet Network (OPN), hosted by the United Nations Environment Programme (UNEP), have collaboratively embarked on developing a Global Circularity Protocol (GCP) for Business. The GCP aims to be the go-to action framework to guide companies in target setting, measuring, reporting and disclosing progress on resource efficiency and circularity to accelerate the shift to circular business models and a regenerative economy.

While data underpins both the CTI and GCP, data sharing faces significant obstacles, including technical, security and organizational issues. The obstacles, such as limited interoperability between systems, hinder the smooth transfer of data across platforms. Furthermore, concerns over data confidentiality – particularly the need to protect sensitive information and intellectual property – make companies hesitate to share data openly. High implementation costs and complex requirements for data-sharing protocols add further deterrents. Without standardized frameworks, these obstacles contribute to fragmentation and reduce the overall effectiveness of cross-sector data-sharing efforts.

To address these barriers, we reviewed leading initiatives in the data-exchange space, including WBCSD's Partnership for Carbon Transparency (PACT) approach, to propose the following high-level recommendations:

- Unifying circularity data definitions and information requirements for data categories in all frameworks, enabling consistent references across different sectors. This will enable more opportunities to apply different frameworks based on the scope and goal of the assessment.
- Linking circularity data with impact data – such as carbon reductions – to support the adoption and scale of exchanging data related to circularity.
- Developing a data model that can accommodate products designed for various circular strategies, along with regular data updates, to support the dynamic nature of circularity data.
- Extending PACT's trust mechanisms related to chain of custody and the verification or audible nature of these to the context of circular data to ensure the accuracy and credibility of circularity data and provide companies with the confidence to share information transparently. To support this, a set of data exchange protocols can safeguard the sharing of confidential information among different participants in the value chain.

By establishing verification mechanisms and data integrity standards, the PACT approach offers a pathway to achieving consistent, credible circularity data reporting, enhancing transparency and stakeholder trust in circular initiatives.

CDX will create the enabling environment to support the GCP for companies to exchange consistent, comparable and credible circularity data at the product level, utilizing the findings from this paper. Launched in January 2024, the scoping phase of the Circularity Data Exchange (CDX) project has explored existing challenges and potential solutions for accounting[1] and exchanging circularity data.

The next steps to create a cohesive circularity data ecosystem include priority actions such as establishing a standardized core set of circularity data points applicable across industries, facilitating the integration of industry-specific standards and advancing technological enablers to support seamless data exchange. Additionally, building collaborative frameworks with key stakeholders and fostering financial incentives will drive broader adoption and ensure that circularity data collection and exchange practices align with regulatory and sustainability goals.

Prioritizing these next phases will establish a foundation for consistent circularity measurement, enhancing comparability across industries and supporting the long-term success of a circular economy.

1. Introduction

1.1 The role of data exchange in a circular economy

Growing policies, reporting requirements, and business focus on circularity

The circular economy is an opportunity to achieve social, economic and environmental sustainability goals. It directly addresses the interconnected crises of climate change, nature and biodiversity loss, and pollution and waste, offering a pathway to mitigate these global challenges. As only 7.2% of the global economy operates on circular principles, there is much room for improvement.[2] By redesigning products, reducing resource extraction and minimizing waste, the circular economy contributes to achieving net-zero goals and supports sustainable development through economic innovation, green job creation and growing regulatory push to embed these practices further.[3] Nearly 3,000 new circular economy commitments, including 135 policies, are under development in 17 sectors worldwide. Their aim is to reduce waste generation and raw material resource extraction and foster informed purchasing decisions and sustainable production.[4]

These policies are creating an urgent need for companies to implement circular practices, emphasizing the need for high-quality data that can serve various purposes relevant to transitioning to and operating a circular economy, such as:

- **Regulatory compliance –** aligning with emerging policies that mandate transparency and reporting circularity-related data;
- **Performance measurement –** assessments that require consistent and comprehensive data to benchmark and monitor circularity metrics over time and help inform target setting;
- **Internal reporting –** communicating circularity through sustainability reports, enhancing stakeholder communication and informing strategy

The importance and the exchange of product circularity data

Most products are designed today for a linear economy, whereby resources are extracted, and transformed into products for consumers who use and ultimately discard them – often ending up in landfills or incinerators. These products rarely undergo processes like repair, reuse or remanufacturing that could extend their lifespan, resulting in missed value-capture opportunities before end-of-life recovery. This approach leads to the waste of materials and restricts their reintegration into the economy.[5] In contrast, the design of a circular economy eliminates waste by maintaining products and materials in circulation at their highest value. This involves employing R-strategies (refuse, rethink, reduce, reuse, repair, refurbish, remanufacture, repurpose, recycle and recover), extending material use cycles and ensuring that end-of-use material recovery returns resources back into industrial systems for the production of new products (for products in the technosphere) or broken down and returned to the Earth (for products in the biosphere).

Product circularity data is critical to understanding the content and composition of a product, as well as the fate of that product – both in theory and in practice. It includes information on the circular aspects of a product and can serve to enable the production of more sustainable and circular products.[6] With product circularity data, companies can optimize resource use and improve their "ecodesign", adapting business models to ensure materials and products stay in the economy at their highest value.

Product circularity data

Other stakeholders could use the standardized information on the circularity aspects of a product used partially or entirely to enable circular evaluation of the product (adapted from the ISO Product Circularity Data Sheet (PCDS 59040). For this reason, product circularity data can be information collected and measured.

This data can be used for:

- **Product design and innovation –** leveraging historical data about products to inform product design improvements and develop alternative product models;
- **Maintenance and service optimization –** using history and wear-and-tear data to enhance maintenance schedules to optimize performance and extend the function life of products;
- **Recovery processes –** using diagnostic data to improve efficiencies, reduce waste and save resources at the end of a product's life;
- **End-of-life management –** providing sorters and recyclers with product disassembly information to improve the quality of material streams, ensuring better recovery rates and resource integration.

An effective circular economy collects and shares comprehensive data about a product and its life cycles at every stage of the value chain to maintain high-quality material loops. This means a continuous circulation of resources and the exchange of information.

Data for circularity performance measurement

Recognizing the need for companies to measure their progress towards a circular economy, WBCSD has developed the Circular Transition Indicators (CTI) – a universal framework of metrics to measure circular performance. Now in its fourth edition since its first release in 2020, CTI helps companies develop insights into how to increase circularity along their value chain and understand how to reduce impacts on climate and nature.[7]

Building on the CTI, WBCSD and the One Planet Network (OPN), hosted by the United Nations Environment Programme (UNEP), have collaboratively embarked on developing a Global Circularity Protocol (GCP) for Business. The GCP aims to be the go-to action framework to guide companies in target setting, measuring, reporting and disclosing progress on resource efficiency and circularity. It combines with comprehensive and targeted policy guidance to accelerate the shift to circular business models and a regenerative economy.

Data underpins the CTI and GCP. As identified in the GCP Impact Analysis, "effective data exchange within value chains and networks is vital. The GCP can aid this by providing standardization and data and digitalization guidance to enhance data availability and interoperability."

To support GCP, we have sought to create an enabling environment for companies to exchange consistent, comparable and credible circularity data at the product level. Launched in January 2024, the scoping phase of the Circularity Data Exchange (CDX) project has explored existing challenges and potential solutions for accounting and exchanging circularity data, with the aim of informing the development of the CDX. [8] This white paper presents the results of those findings.

Circular accounting

The practice of measuring and analyzing a company's financial and non-financial performance to truly reflect and account for the value and impact of circular businesses.

Note: this paper covers only non-financial performance.

1.2 Scope of this paper

Data is necessary to inform decision-making, develop and adopt circular business models and measure company performance. To that end, companies need access to consistent, comparable and credible circularity data at the product level. However, there are currently limits to information collection and sharing. Challenges exist around data availability, accessibility, reliability and comparability.[9] The lack of quality shared data presents a significant roadblock for businesses aspiring to adopt circular business models and set and meet their circular targets.

How can businesses facilitate effective product circularity data exchange?

To answer this question, it is crucial to understand the scope of product circularity data and determine what data to report. We have taken the following approach:

1. A comprehensive landscape review outlining the current regulations, voluntary frameworks and leading data initiatives, including a mapping of these frameworks/initiatives across the value chain;
2. A deep dive into the processes, tools and challenges faced by companies in exchanging product circularity data along the value chain by stakeholder interviews and analyzing case study examples;
3. Providing recommendations on how WBCSD can support product circularity data accounting and exchange, including insights and best practices from WBCSD's Partnership for Carbon Transparency (PACT) applicable to the context of circularity data (Box *1*).

Box 1: Learning from product carbon footprint data exchange

To guide the approach for circularity data exchange, this paper includes a review of WBCSD's Partnership on Carbon Transparency (PACT), which has established a standard for scope 3 carbon emissions accounting and data exchange at the product level. This paper explores whether and how it is possible to adapt PACT's approach to product carbon footprint (PCF) data – encompassing methodology (calculation, data integrity, assurance and verification), cradle-to-gate PCF data exchange, and technical specifications (data model and application programming interface (API)) – for product circularity.

The PACT framework, detailed further in the section on learning from PCF exchange, offers lessons that could inform the development of circularity data exchange (CDX) moving forward.

2. Defining and measuring product circularity

2.1. Key data categories

An analysis of existing regulations and frameworks (see Appendix *1*) offered a glimpse into how comprehensively they support realizing the circular economy's potential and what it means for companies in terms of the data they need to collect to meet their requirements.

The analysis identified and clustered the information requirements related to circularity into **14 data categories** as listed in Table *1*. *(Note: This categorization aligns with those presented in leading circularity initiatives, such as the European Union's Ecodesign for Sustainable Products Regulation (ESPR),[10] the International Organization for Standardization (ISO) Product Circularity Data Sheet[11] and the Battery Passport.[12])*

The analysis showed the data currently requested by regulatory frameworks and voluntary initiatives falls predominantly in three categories:

- **Reuse and recycling information;**
- **Materials and chemical substances;**
- **Environmental information.**

These categories represent only part of the breadth of data required to support a circular economy, in which reuse data, for example, would not be a priority over product identification data, as both play a pivotal role in the circular paradigm. The significant focus on recycling information – one R-strategy considered less effective in retaining material value than other R-strategies – highlights the dominance of the linear-economy mindset across frameworks and initiatives. Complementary information, such as product design data that facilitates repair, remanufacturing and disassembly, receives comparatively less attention in the existing regulatory and voluntary frameworks analyzed, though they are beneficial to extending product lifetime.

Additionally, while currently absent from current data requirements from regulatory frameworks and voluntary initiatives, several relevant areas of information may also prove essential in enabling and operating of circular business models. For example:

- **Product authentication data and certifications** may prove relevant for maintaining the reuse and resale values of products as they enter a second-life phase, and beyond.
- Additional information on **materials and their origins** may need incorporation, especially in the case of products that use bio-based materials (for example, 60% bio-based polymer derived from corn starch). The renewability of materials (such as sourced from Forest Stewardship Council (FSC)-certified forests), as well as end-of-life guidance (compostable packaging requires industrial composting facilities) may prove essential in reaping the benefits of the bio-based nature of these materials over non-biobased alternatives.
- Information connected to other relevant regulations or **product-specific directives** (like packaging and packaging waste) may also streamline processes for handling, disposal and recovery, as well as enhance alignment with evolving regulatory requirements.

While organizations already collect much of this information, it is not often accessible across the value chain. In some cases, product-level information collected at one stage of the value chain may inform decision-making at "higher" levels of the value chain. For instance, records of **product history data**, such as wear-and-tear information collected from a product's use phase, can provide brands with information on designing out faulty aspects of a product that limit product lifetime. In such cases, data must be collected at the consumer-level and be receivable at the brand/manufacturing level, in order to track the condition of the product. This case highlights the dynamic nature of the data needed to inform circular practices and the maturing/evolving space required to host it – beyond the current frameworks requesting and mandating this data.

Table 1: 14 data categories identified as relevant for the pre-use phase, use phase and end of life of a product and additional information.

Pre use phase	
1	**Product identification**: Information on the product's identifying characteristics, which is necessary to link the physical product with the product-related information. *Example: iPhone 12 Pro Max, Model A2411, Serial #GH45K92L, 128GB, Pacific Blue*
2	**Materials**: Information on material characteristics used in the life cycle of the product. This includes information on the origins of the materials, the presence of critical materials and material flows such as % of recovered materials. *Example: % recycled content, Monkey Glass front, lithium-ion battery, reused aluminum casing, FSC certified paper.*
3.	**Design for better use and disassembly**: Data on how the company's product design facilitates better use and easy disassembly at its end of life (EOL). This includes design features that enable, for example, repairability. *Example: Modular design with replaceable components, use of standard screws, easy-to-separate material design.*
4.	**Technical specifications**: Details of the product's performance and technical capabilities, including power ratings, dimensions, weight and other relevant specifications. *Example: 6.7" OLED display, B11 Bionic chip, 5G capable, triple 12MP cameras, IP69 water resistance*
5.	**Product life cycle**: Information on the consecutive, interlinked stages of a product, from raw material use to final disposal. This information can inform the environmental impact of a product (such as life-cycle analyses). *Example: Raw material extraction, emission from transportation/distribution, 5-year usage, recyclable disposal*
6.	**Chemical substances**: Information on the chemical substances present in the product, including the presence of hazardous materials, compliance with chemical regulations and safe handling instructions. *Example: Contains lithium, cobalt, nickel; complies with REACH regulations; safe handling instructions included*
Use Phase	
7.	**Product maintenance**: Records of wear, product service, repairs and upgrades performed as well as any maintenance schedules. *Example: Annual servicing records, battery replacement after 2 years, available firmware updates logged.*
8.	**Warranty information**: Details of the product's warranty coverage, including any applicable terms and conditions. *Example: 2-year limited warranty covering manufacturing defects but excluding accidental damage, with registration required within 30 days of purchase.*
9.	**User manuals**: Instructions on how to properly use and maintain the product safely and effectively. *Example: Step-by-step setup guide, troubleshooting tips for common issues, maintenance schedule and locations, safety warnings, and disposal instructions.*
End-of-Life Phase	
10.	**Reuse and recycling information**: Information to inform the appropriate end-of-life management. For a product being resold, this may include information on wear-and-tear to inform repairs/maintenance for resale. For recycling, this information may cover material makeup. *Example: % recyclable materials; retail location for repair and resale; packaging materials that can be recycled*
11.	**Energy recovery**: Information to inform energy generation potential through direct incineration with the recovery of heat through combustible waste. *Example: Waste parts not suitable for recycling can be incinerated in specialized facilities to generate heat or electricity, contributing to the reduction of fossil fuel use.*
12.	**Waste**: Any substance, material or object listed in the categories set out in Annex I to Directive 2006/12/EC created during the product's production, or in which the holder discards. *Example: Used batteries classified under EWC (European Waste Catalogue) code 16 06 05* as chemical waste, indicating the need for special disposal methods to avoid environmental harm.*
13.	**Hazardous waste management**: Any categories or types of waste in liquid, sludge or solid form listed in Council Directive 91/689/EEC according to their nature or the activity which generated them. *Example: Spent fluorescent tubes containing mercury, categorized under EWC code 20 01 21*, requiring careful handling and disposal to prevent mercury release into the environment*
Additional Information	
14.	**Environmental Information**: Information on whether the product and its parts can interact with or change the environment wholly or partially at any stage during the product's life cycle. *Example: use of critical raw materials, possible adverse effects if enters the environments, noise level, emissions data.*

The analysis also showed the data categories organizations most sought after along the value chain (see Box 2 for a visualization). For example, of the regulations and frameworks considered, most operate at the design and production stages and request information from various data categories – both of which diminish further down the value chain; only one regulation tackled the raw material sourcing and procurement stage, where *environmental information* and *materials* data categories are relevant. The only frameworks/regulations requesting life-cycle data were those that took a value-chain approach.

It is important to note that a greater number of regulations/frameworks across a value-chain position does not equate to greater coverage of circularity data and in fact could result in the opposite. The current landscape suggests the dispersion and limited harmonization of frameworks whereby the number of existing and emerging frameworks adds complexity to an already complex space.

Box 2: A visualization of regulations and frameworks mapped along their respective value chains and the data categories sought after

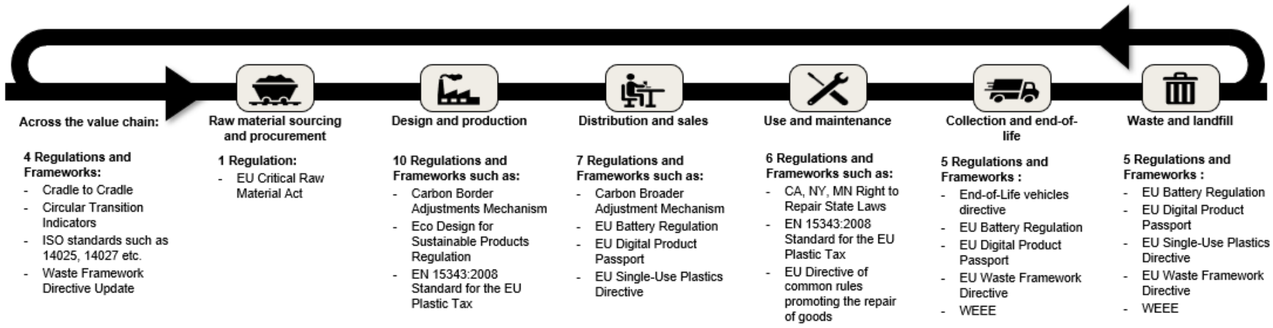


Figure 1: Regulations and voluntary initiatives mapped across their respective interventions, across the value chain (Source: authors of this paper).

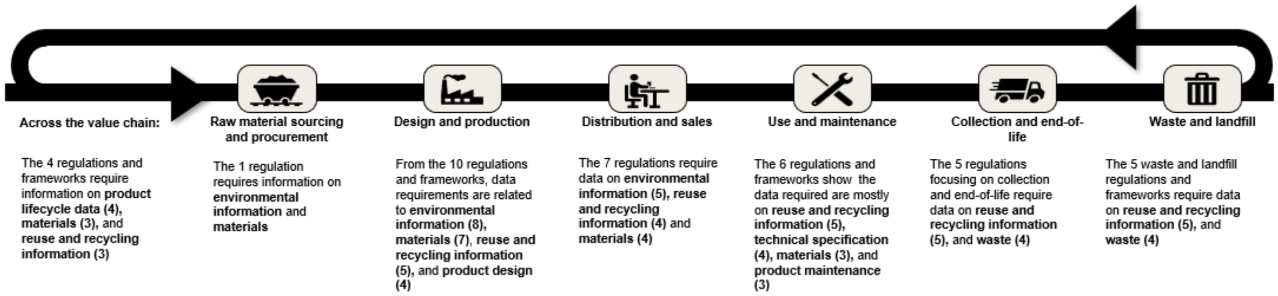


Figure 2: Data categories sought after by regulatory frameworks and voluntary initiatives mapped along the value chain Source: authors of this paper).

2.2. Challenges hindering defining and harmonizing circularity data

2.2.1 Defining "Product circularity data"

The analysis of existing regulations and frameworks revealed that product circularity data is **not limited to a single data point**, as is often the case with carbon emissions, but rather encompasses a broad range of data categories. Each category provides a unique perspective on product circularity, offering valuable insights that can support the development and implementation of circular business models. Understanding how the circularity potential/value of materials and products increases or decreases at each stage of the value chain requires multiple data points in each data category.

Comparing the content of the information sought after under each category revealed inconsistencies in the definition of product circularity.

For example, while the Battery Passport and the ISO Product Circularity Data Sheet align on the content of data such as *general product*, *company information* and *material composition of hazardous substances*, though the same is not true for the Ecodesign for Sustainability Products Regulation's (ESPR) ecodesign requirements, which is largely a result of the scope of these initiatives. The Battery Passport takes a product-specific approach to defining circularity data and prioritizes information on material composition and technical performance metrics – like battery chemical composition, capacity and energy efficiency – focusing on how the product performs and lasts in operational terms. In contrast, the ISO Product Circularity Data Sheet,[13] which takes a product-agnostic approach, emphasizes design features that support product longevity, reusability and recyclability, yet lacks in-depth technical performance information specific to the product level. In the context of the ESPR, Digital Product Passports (DPP) will contain product group-specific criteria and thresholds, tailoring requirements to the characteristics of each product group (see Appendix *2*).

In the absence of sector-agnostic definitions for foundational circularity data, on which sectors build their specific additions, it is easy to create fragmentation in the use and understanding of circularity data. This lack of consensus between initiatives produces inefficiencies for companies as they must align with the multitude of definitions of circularity, increasing the gap in enabling cross-sector collaboration.

2.2.2 Operationalizing product circularity data

In parallel, discrepancies arise in how initiatives **operationalize circularity data** relative to the R-strategies (see Appendix *3*). For example, frameworks interpret "reuse" differently, which results in inconsistencies across data reporting requirements. Table *2* highlights the variations in terms between sector-agnostic and sector-specific frameworks in different regulations and frameworks.

Table 2: An example of the discrepancies observed across definitions of five of the R-strategies

	European Commission	Circular Transition Indicators	Waste Electrical and Electronic Equipment (WEEE) Directive	ISO 59004	European Committee for Electrotechnical Standardization (CENELEC) 45553
Reuse	Re-use of a product which is still in good condition and **fulfils its original function** (and is not waste) for the **same purpose for which it was conceived**	To extend a product's lifetime beyond its intentional designed life span, **without changes made** to the product or its functionality	Any operation by which products or components that are not waste are used again for the **same purpose** for which they were conceived	Use of a product after its initial use, for the **same purpose** for which it was originally designed	Process by which a product or its parts, having reached the end of their first use, are used for **the same purpose** for which they were conceived
Repair	Repair and maintenance of defective product so it can be used with its **original function**	To extend a product's lifetime **by resorting** it after breakage or tearing **without changes made** to the product or its functionality	'Preparing for re-use' means checking, cleaning or repairing recovery operations, by which products or components of products that have become waste are prepared so that they can be **re-used without any other pre-processing**	Action to **restore** a product to a condition needed for the product to function according to **its original purpose**	Process of **returning** a faulty product to a condition where it can fulfil its **intended use**
Remanu-facture	Use parts of a discarded product in a **new product with the same function** (and as-new-condition)	**To disassemble** a product to the component level and reassemble to (replacing components where necessary) **as-new condition with possible changes made to the functionality** of the product	N.a.	**Return** an item to **original condition** from both a quality and performance perspective **using an industrial process**	**Industrial process** which produces a product from used products or used parts where at least one change is made **which influences the safety, original performance, purpose or type of product**
Refurbish	Restore an old product and **bring it up to date** (to specified quality level)	To **extend a product's lifetime** by large repair, potentially with replacement of parts, **without changes made to the product's function-ality**	N.a.	**Restore an item**, during its expected service life, to a useful condition for the **same purpose** with at least similar quality and performance characteristics.	**Industrial process** which produces a product from used products or used parts where **at least one change is made**
Recycle	Recover materials from waste to be **reprocessed into new products, materials or substances whether for the original or other purposes**. It includes the reprocessing of organic material but does not include energy recovery and the reprocessing into materials that are to be used as fuels or for backfilling operations	To reduce a product back to its material level, thereby allowing the **use of those materials in new products**	**Any recovery operation** by which waste materials are reprocessed into products, materials or substances whether for the original or other purposes.	**Activities to obtain recovered resources** for use in a process or a product, excluding energy recovery	N.a.

The variation in these definitions results in changing data requirements under their respective data categories. This means some areas are more or less represented than others, depending on the framework requiring this data. The use of this data for circularity measurements exacerbates the absence of unified definitions, meaning the methodologies behind voluntary circularity frameworks being adopted typically differ.

For example, while both the Circular Transition Indicators (CTI) and Material Circularity Indicator (MCI) aim to measure circularity (performance), they use differing formulas based on their definitions of material circularity. As CTI is flow-based, it defines material circularity as the weighted average of circular inflow (recycled or renewable inputs) and circular outflow (recycled or renewable outputs). MCI on the other hand considers the functional life cycle of a product in its definition and so also factors in the product utility function, such as intensity of use, to account for the effective use of materials over the product's life cycle.[14]

While different frameworks can offer complementary perspectives on circularity, these different approaches limit the comparability of results. Discrepancies in the definitions used across these frameworks have also led **to inconsistent implementation within and across industries**, whereby data interpretation creates greater divergence with circularity data accounting.

2.2.3 Data availability at the product level

Finally, **reliable, primary product-level data**, which is critical in making accurate and informed decisions, **is largely unavailable**. This is typically a result of data not being collected; data not being stored in a centralised location internally, or not accessible through the value chain. In its absence, organizations many rely on secondary data, such as regional or industry averages, which often fails to reflect true values. For instance, a company might use regional recycling rates for a material in its product, overlooking design factors that limit actual recyclability. This leads to inaccurate measurements and misses opportunities for product and process improvements.

Product-level primary data collection and tracking is often complex, particularly in recycled content verification or tracking waste valorization. Many producers depend on procurement data, which typically fails to capture critical metrics like carbon footprint or end-of-life recycling rates. ‘Upstream’ actors, such as brands and manufacturers, often struggle to gather reliable data beyond tier 1 suppliers, risking financial impacts when tracking critical materials or entering new markets. ‘Downstream’ actors face similar challenges, lacking visibility into a product’s post-consumer fate – whether reused, recycled or landfilled – which is increasingly vital under extended producer responsibility (EPR) schemes that demand proof of proper recovery and processing. Without this, they will face fees.

Furthermore, this gap in data collection and availability hinders companies from collecting and processing data to make informed decisions, implement circularity strategies and benefit from optimizing material use accordingly.

However, regulations like the upcoming Corporate Sustainability Reporting Directive (CSRD)[15] are likely to drive more comprehensive data collection in the near future, which will push companies to improve transparency and accountability.

3. Exchanging circularity data

3.1. Barriers

Having explored the challenges of defining product circularity data, the question of how industries share this data still stands. The seamless exchange of circularity data across value chains is critical to unlocking the full potential of a circular economy, though it brings additional insights and challenges, both on a technical and organizational level. Interviews with industry stakeholders and real-world use cases reveal how complexities in technology, data confidentiality and alignment across diverse actors create significant barriers in data exchange at the product level.

One of the key challenges in exchanging product-level data lies in the interplay **between information technology (IT) complexities and data confidentiality concerns**. Technical barriers, such as the lack of interoperable tools, hinder seamless data sharing across systems and platforms while maintaining an auditable chain of custody. Additionally, apprehensions about protecting sensitive information – especially intellectual property (IP) – discourage companies from sharing data without robust verification mechanisms. The high costs of implementing secure data-sharing systems compound this hesitation, as does the significant technical infrastructure required to support them effectively.

A further barrier to adopting circular models is **the absence or lack of proven, scalable business incentives**. Many companies struggle to justify investments in the collection of product circularity data, particularly when the benefits associated with the data are unevenly distributed among the actors involved in the collection process. The perceived value of detailed circularity data is not always clear, making it challenging to demonstrate a strong business case. This creates a “chicken and egg” dilemma where companies hesitate to invest without tangible returns, stalling progress on the broader adoption of circular practices.

Moreover, there is a tendency for **climate change** initiatives **to overshadow circularity efforts**, as the former tends to take precedence in corporate sustainability agendas. This imbalance results in limited attention and resources allocated to circularity data collection. However, this situation reveals a paradox: although climate action is the priority, the **circular economy is a vital component in addressing broader climate and environmental impacts.** Beyond reducing greenhouse gas emissions, circular strategies have the potential to address unsustainable consumption and production, lower pollution, improve land- and water-use efficiency and reduce overall resource extraction, particularly in high-impact areas (like cement, steel, plastics and aluminum). Obtaining data to enable informed circular practices is therefore fundamental to climate action.

Lastly, collaboration across supply chains also presents its own challenges, particularly in establishing common data-sharing practices. The complexity arises from **aligning disparate systems, tools and regulatory frameworks across actors and geographies**. To advance circularity effectively, companies must explore collaborative models and foster cross-industry synergies that facilitate a unified approach to data exchange. Only through enhanced cooperation and standardization can the seamless exchange of circularity data become a reality, ultimately enabling a more circular economy.

3.2 Learning from leading data exchange initiatives

Several initiatives have emerged to address the data-sharing needs of both sector-specific and sector-agnostic ecosystems (see Appendix *4*). As unlocking circularity data exchange is at an early stage, these lessons learnt are extracted from leading data initiatives, whose primary aim/scope may not be enabling circularity. Nevertheless, these initiatives can provide important insights into the technical, organizational, and security-related barriers that companies face in enabling data exchange and be further built upon to provide a better understanding of existing gaps and opportunities to create a unified approach to enable credible product circularity data exchange.

Three key insights that emerged and that the analysis highlighted as important areas to build on in the context of circularity include:

1. Many data initiatives have a primary focus on managing and accessing data, often neglecting other crucial aspects like the quality of data (such as primary versus secondary data) and traceability of environmental product information. This oversight can result in the exchange and use of incorrect data, creating challenges in making accurately informed decisions and promoting circular practices. Information flows through the value chain further exacerbate this, with every input of inaccurate information moving further from the “truth”.
2. While several initiatives emphasize the need for interoperability, many have not fully matured in practice. Efforts to standardize data exchange are evident but the lack of cross-sector initiatives hinders global data exchange frameworks.
3. Most initiatives are still in the implementation stage, with only a few at scale. This diversity in maturity levels presents both opportunities for growth and innovation, as well as challenges in achieving interoperability at scale with existing misaligned initiatives.

As the development of circularity data exchange is still nascent, different sectors are adopting varying approaches to collaboration and data sharing. The pyramid model in Figure *3* illustrates this spectrum for a few of the leading initiatives, frameworks and standards considered in the landscape analysis. These range from sector-specific initiatives to sector-agnostic solutions and from alliances focused on coordination within sectors to interoperable data spaces that enable cross-sectoral exchange.

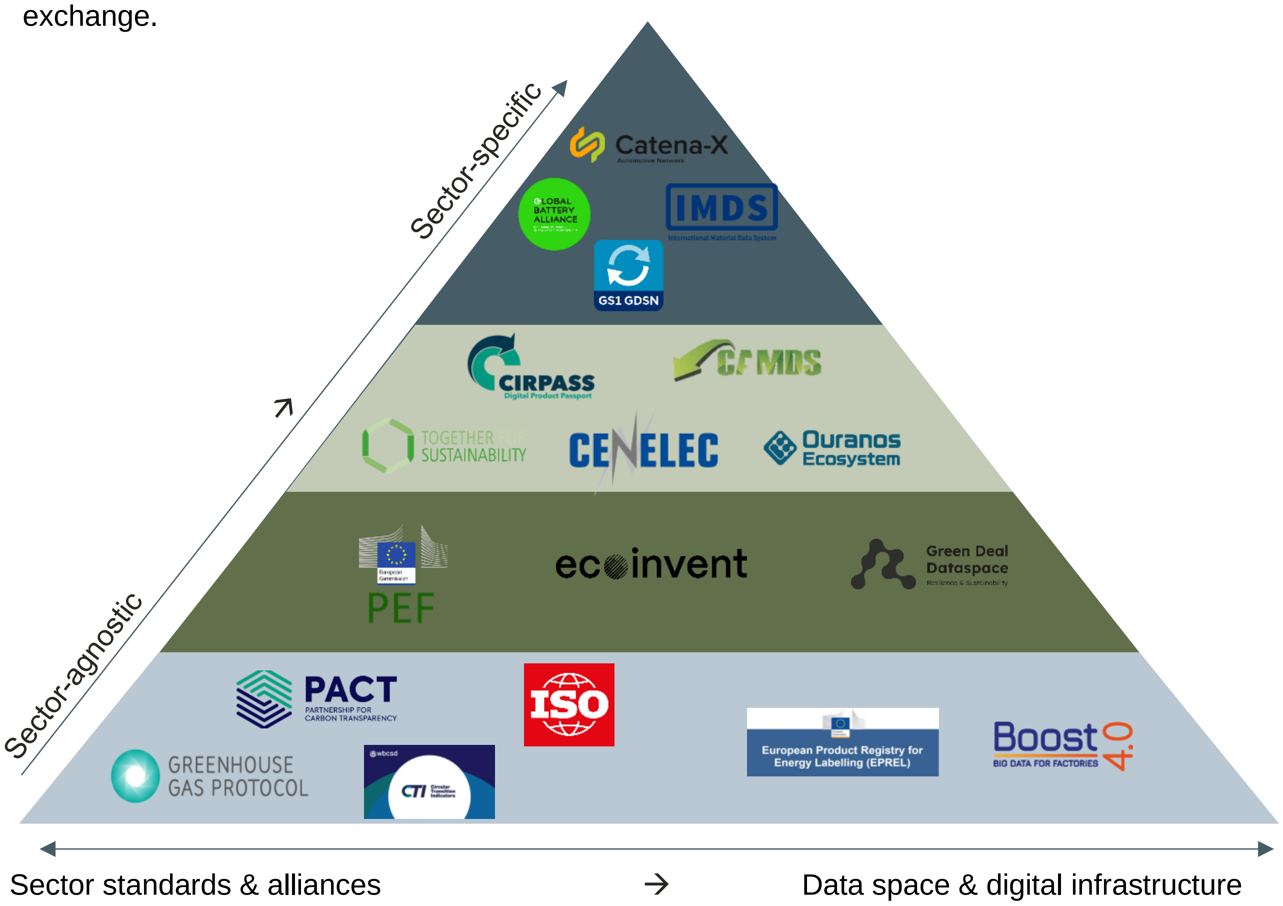


Figure 3: A pyramid model illustrating sector-agnostic to sector-specific approaches and the shift from sector standards and alliances to interoperable data spaces.

Sector-agnostic initiatives are at the base of the pyramid. They provide fundamental data categories relevant across sectors. Product-level initiatives, such as PACT PCF exchange (see the section on learning from PCF exchange), may serve as the groundwork for the exchange of product-level circularity data, as their design creates a broad foundation for and across industries.

Moving up the pyramid and building on this foundation, **sector-specific initiatives** emerge that address the unique aspects of the industries. The automotive and chemical industries, for example, have long-established alliances and databases, such as International Material Data System (IMDS), that facilitate the sharing of material data. In these sectors, while nascent, the exchange of data has matured more rapidly thanks to frameworks such as the PACT Technical Specifications, which presents a common standard for interoperable data exchange.

However, adapting these frameworks to meet sector-specific needs closely links to hashing out **challenges related to the respective sectors and their products, often stretching along the value chain**. For example, in the automotive sector, companies that need to consider vehicle life-cycle management and the use of recovered materials (required under the Directive on end-of-life vehicles[16] (ELV Directive)) are reliant on the exchange of information from end-of-life recovery actors to enable end-to-end material tracking. Such tracking can show material flows and impacts from production through to end-of-life recycling and back into production. The success of these initiatives thus also rides on cross-value chain collaboration. The Global Battery Alliance (see Box *3*) is an example of a leading initiative enabling information exchange in the automotive sector as a result of increasing regulatory demands for battery information.[17] [18]

Box 3: Global Battery Alliance

The **Global Battery Alliance (GBA)** is a multi-stakeholder initiative that unites organizations across the battery value chain to enhance **supply chain transparency.** To achieve this, the GBA developed the **Battery Passport** concept, aligned with the EU's Ecodesign for Sustainable Products Regulations (ESPR) and the EU Battery Regulation. The Battery Passport serves as a scalable solution to standardize circularity data sharing within the battery industry. It is currently being operationalized through the **Battery Pass** (Figure *4)*. To date, the Battery Passport has undergone two piloting phases, involving original equipment manufacturers (OEMs), representing 80% of the electric vehicle market, and major upstream cobalt producers, accounting for the top three global cobalt producers as of 2022.

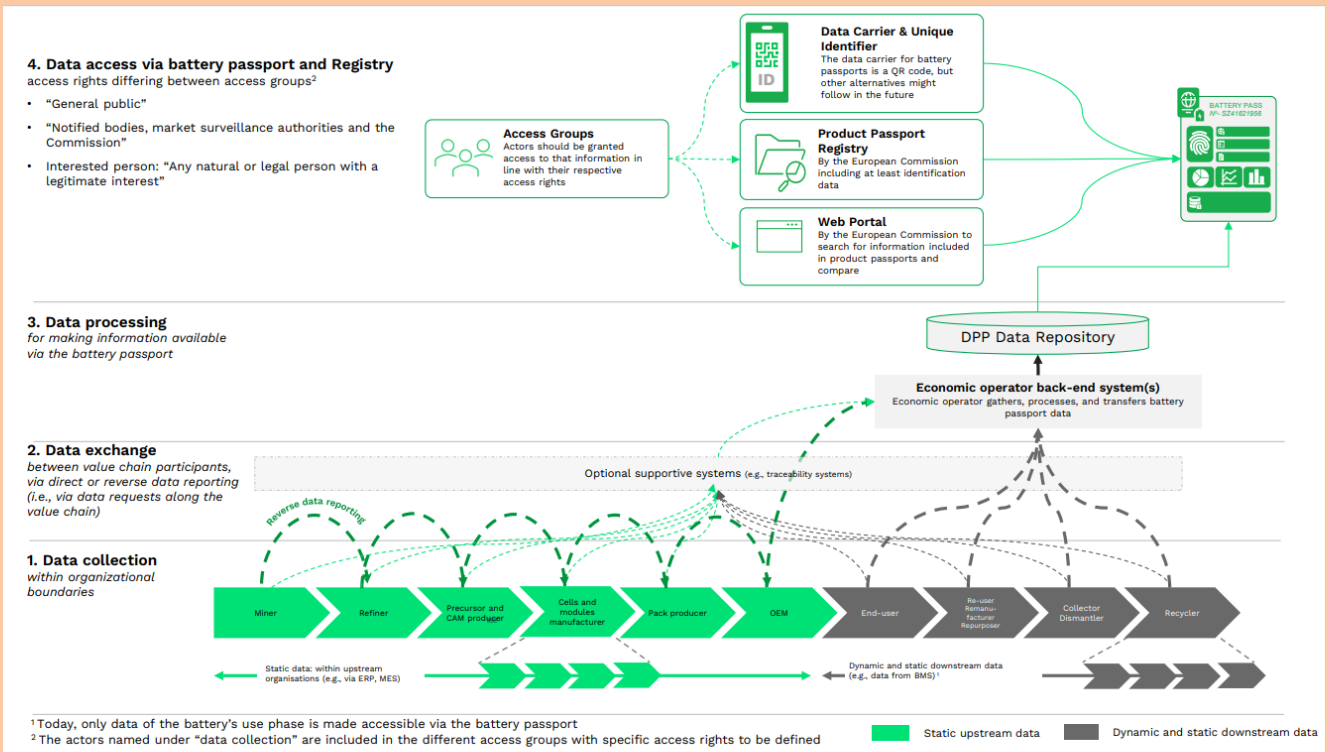


Figure 4: Technical design and operation of the Global Battery Alliance's battery passport (Battery Pass, 2023).

The design of the Battery Passport meets the EU Battery Regulation's requirement for economic operators to establish **traceability systems** that ensure transparency throughout the entire value chain. Implementing such a system requires significant **data consolidation** across multiple value chain participants and substantial infrastructure investment. By leveraging the collection of mandatory material reporting data – as well as voluntary value chain assurances or provenance data – the Battery Passport has the potential to **create a reliable record of materials**, tracking their journey both upstream and downstream.

The **Battery Passport's technical standard stack** provides guidelines for the IT infrastructure and domain data ecosystem needed to ensure effective data collection, exchange, processing and reporting. The EU Battery Regulation emphasizes the need for primary/company-specific data for materials and components. As companies increasingly prioritize collecting primary data, the shift will improve differentiation and optimization across entire value chains. Over time, upstream data aggregation and exchange – where each actor adds scope 1 and scope 2 emissions data to received materials and components – will provide increasingly thorough, reliable and accurate insights into material flows and their impacts.

While companies will likely not share sensitive data due to confidentiality, the sharing of aggregated data, like product carbon footprints, does not face the same concerns.

The design of the **standard stack** targets upstream data aggregation for this type of data exchange, as envisioned by the Global Battery Alliance and other leading initiatives, such as Catena-X and PACT.

Reflecting on these initiatives provides areas for consideration to support and further advance circularity data exchange. The Battery Passport pilots, for example, underscore the potential for traceability systems to serve as the backbone of circularity data exchange. It has demonstrated granular tracking of material flows, which could serve as a blueprint for broader adoption across industries and to create value for the business ecosystem.

Other areas include the need for interoperability between systems, ensuring compatibility and secure intra-company data collection and sharing. Designing standardized interfaces to track materials and products will be essential, as will the establishment of common identification and data exchange standards. For instance, Gaia-X's technical framework offers a promising starting point in creating interoperable and secure data-sharing ecosystems.[19]

Moreover, new business models could emerge from product and material tracking tools. These systems will need to address both technical challenges and evolving business needs to unlock the potential for businesses to optimize operations, meet regulatory requirements and fully embrace circular economy principles.

4. Learning from product carbon footprint exchange

Hosted by WBCSD, the Partnership for Carbon Transparency (PACT) tackles one of the major obstacles experienced in advancing a circular economy: standardized data exchange. Taken together, the PACT Methodology, Network and Technical Specifications facilitate the accounting and exchange of product carbon footprint (PCF) data across industries. Analyzing the PACT Methodology and PACT Solutions (meaning PACT-conformant companies) may reveal opportunities to leverage this framework in the context of circularity.

4.1 The PACT Methodology

A core element of the PACT Methodology is the **standardization of scope 3 carbon emissions accounting** at the product level, building on established guidelines like the Greenhouse Gas (GHG) Protocol and ISO 14067. This is essential in promoting a unified understanding of carbon emissions data and allows companies to effectively track emissions, further enabling them to act on this data (meaning reducing PCF values).

The methodology emphasizes **data integrity** and the need for **assurance and verification**. It supports this by providing:

- Guidance on how to increase the share of quality primary data used to calculate PCFs;
- Guidance on how to select secondary data in the absence of primary data;
- The metrics to track, report and improve data quality, and increase the integrity and consistency of carbon emission data.

Further direction on how to define requirements for PCF assurance and verification creates an enabling environment to ensure the reliability of the emissions data – which is also necessary to undergo a carbon audit – and create trust among stakeholders. This direction includes considerations for the scoping and rollout of assurance and verification requirements across time horizons, the provision of standardized and relevant evidence to substantiate emissions claims, as well as additional information on supporting small and medium-sized enterprises (SMEs) and on process and reporting, operationalizing the guidance for its users.

The methodology also touches on **data exchange guidelines**. They indicate the minimum information requirements companies must exchange in addition to the PCF, such as product information (like a data provider's company name) and data reliability information (the share of primary data in a PCF, for instance), leading to greater credibility. The data exchange guidelines connect to technical specifications that PACT-conformant solution providers use to enable the transparent, straightforward and confidential exchange of emissions data across value chains and industries.

4.2 PACT Network and Technical Specifications

The PACT Network is an open and global network of interoperable solutions to enable the secure peer-to-peer exchange of accurate, primary and verified product emissions data, across all industries and value chains. By focusing on peer-to-peer connections – to protect sensitive information – the PACT Network can ensure data confidentiality. This encourages data sharing without risking privacy breaches and helps foster collaboration and confidence in carbon footprint data management.

PACT Solutions on the PACT Network must implement the PACT Technical Specifications, which certify that they are interoperable with each other. The Technical Specifications articulate the data semantics and standards that allow companies to build application programming interfaces (APIs) that are interoperable with other solutions in the PACT Network. The Technical Specifications also offer data model extensions, which can allow PACT Solutions to provide information beyond PCF data.

The effectiveness of the PACT Methodology, Network and Technical Specifications as an enabler of product carbon footprint data emissions accounting and exchange is evident in its market adoption. Since its development in 2021, companies using the framework have reported improved data accuracy and consistency, which are critical to developing emissions reduction strategies. Having access to detailed carbon footprint data at the product level, shared across partners, has increased transparency and, with that, the facilitation of collaborative efforts to decarbonize initiatives throughout the value chain.

4.3 Adapting PACT for Circularity Data Exchange

While the PACT Methodology and Technical Specifications excel in standardizing carbon footprint data accounting and exchange, there are additional considerations that would benefit extending this to circularity data exchange.

4.3.1 Challenges in adapting PACT for Circularity Data Exchange

The first challenge lies in the **lack of consensus on defining and calculating product circularity data**, such as determining what to measure comes after determining why it is valuable to measure circularity. In the case of carbon emissions and in order to meet carbon reduction targets, there was a need to harmonize product carbon accounting. Thus the PACT Methodology sought to standardize scope 3 carbon accounting. Circularity, on the other hand, is still an evolving space where the “end goal” isn’t as reflective as net-zero targets (for example).

Circularity covers data across various categories, such as material composition, reuse data, functional lifetime extension data, such as durability or recoverability, and recycled content. Despite multiple regulations and frameworks requesting (parts of) this information, no universally agreed upon definition for circularity data, or the information requirements that fall under each data category, exist. As long as frameworks are prescribing concepts like "reuse" in varying ways, the collection of inconsistent data will continue and will complicate efforts to harmonize circularity data across sectors.

Existing circularity assessments, such as CTI, MCI or the European Union’s Product Environmental Footprint (PEF), each have slightly different aims and, thus, scopes. Clarity on the value of exchanging only a specific type of data – for example, the % recycled content of a product – versus additional circularity data to inform company-wide circular strategy greatly determines how broad and granular these data points need to be.

The second challenge concerns the **dynamic nature of circularity data**. Unlike PCF – which applies a static calculation – product circularity data involves tracking changes to a product as it moves through different life-cycle stages. Therefore, each stage or change made requires additions or alterations to a continuously dynamic calculation. For example, a product re-entering the market after refurbishment will need to carry all additional information, differentiating *original* parts from those that are *new*, as well as (where applicable) information on these changes to the product (such as new material makeup, life expectancy, new rules on lifetime extension practices, etc.). This becomes particularly complex when products go through multiple circular strategies operated by different actors at different stages of the product’s life cycle. Accounting for these operations can significantly impact a product’s overall environmental footprint, especially when considering comparisons to a traditional linear product’s life-cycle journey.

4.3.2 Opportunities to adapt PACT for Circularity Data Exchange

While not directly replicable in the context of product circularity data, PACT’s approach does offer valuable insights into creating standardized, secure and interoperable systems for data exchange:

- Drawing on PACT's emphasis on standardization, organizations behind the frameworks could make efforts to unify circularity data definitions and information requirements for data categories in all frameworks, enabling consistent references across different sectors. From there, they would introduce standardized means of collecting this data. And while the results of circularity performance calculations (like CTI, MCI, PEF) will always be incomparable when following differing methodologies, standardizing the elements of the calculations will enable more opportunities to apply different frameworks based on the scope and goal of the assessment.
- Linking circularity data with impact data – such as carbon reductions – may help support the adoption and scale of exchanging data related to circularity.
- Developing a data model that can accommodate products designed for various circular strategies, along with regular data updates, may be key to supporting the dynamic nature of circularity data. Along with this, it is essential to explore the potential of using the extension to the PACT Technical Specification to support this for circularity data.
- Extending PACT's trust mechanisms related to chain of custody and the verification or audible nature of these to the context of circular data can ensure the accuracy and credibility of circularity data and provide companies the confidence to share information transparently. To support this, a set of data exchange protocols can safeguard the sharing of confidential information among different participants in the value chain.

5. Conclusion

The systemic journey towards a fully functional circular economy faces significant obstacles, with **data fragmentation and the lack of standardized circularity definitions as core challenges**. Without clarity on the information needed to enable circularity – and an understanding of the benefits circularity data can bring to decision-making – efforts to collect, share and apply such data effectively will remain inconsistent. Establishing the case for circularity data and addressing confidentiality concerns to build trust among stakeholders and incentivize participation are critical first steps.

While initiatives like PACT provide valuable insights into data management – such as promoting standardization, ensuring data integrity and enabling secure sharing practices – **these efforts alone cannot fully address the complexity of *circularity* data exchange**. The absence of harmonized definitions, the dynamic nature of circularity data (which evolves across life-cycle stages) and the requirement for continuous tracking of products and their materials remain key obstacles. Moreover, technical challenges like IT infrastructure limitations, data confidentiality concerns and the lack of clear business incentives continue to impede progress. Overcoming these challenges will involve more than the adaptation of PACT's Methodology and Technical Specifications. It will require a **comprehensive and coordinated approach that can leverage industry initiatives and frameworks to build a robust foundation**. Efforts must seek to harmonize definitions, design dynamic data models to accommodate evolving product life-cycle stages and implement robust verification mechanisms to ensure data reliability and accuracy.

Insights from leading initiatives such as the Global Battery Alliance, Catena-X and Gaia-X further highlight the **importance of collaboration across sectors.** These examples demonstrate how interoperable tools, digital infrastructure and common data standards create a robust system for seamless data sharing and build trust across value chains. By integrating these lessons, industries can co-create a dynamic system that addresses obstacles and can better anticipate future needs.

5.1 Establishing the foundations for circularity data exchange

Building on the challenges identified, lessons from leading industry initiatives and the landscape of product circularity data requirements by regulatory and voluntary frameworks, the analysis has identified four key areas for further development to create an environment conducive to sharing circularity data.

1. Establishing a core set of circularity data points that are applicable across sectors

Reaching a consensus on standardized data points is crucial to the creation of a unified approach to collecting consistent circularity data. For example, this exploration identified 14 data categories based on the content of existing frameworks and standards, though their information requirements under the different regulations and standards differ. Industries need the harmonization of this process at a global level. This standardized core set of sector-agnostic data requirements will provide the foundation for consistent circularity measurement. This would help to enable the development of data-sharing protocols and, ultimately, the ability to generate comparative results across industries.

2. Tailoring data requirements

Once the core set of sector-agnostic data points become a global standard, the next key area of development would include tailoring data requirements to align with sector-specific attributes at a product or industry level for the appropriate value chain actor**.** This would allow for greater precision in data collection and sharing, as data specific to one product may not be applicable to another. For example, of the data categories relevant to electronic product consumers, access to user manuals and product longevity information would be relevant. For the chemical sector, however, material composition and waste management processes are more relevant to end-of-life value chain actors. Distinguishing these needs is essential for accurate data collection and use (such as communication). Using existing data collected for compliance with mandatory regulation may help lay the groundwork for the further specification of product-specific data requirements.

3. Ensuring product traceability across the value chain

Ensuring product traceability across the value chain may be key to enabling circular models and capturing insights into the dynamic nature of circularity data. As products transition through the value chain and through various life-cycle stages (such as a second life) a shared and consistent record-keeping system is imperative to maintaining the circulation of products in the system at their highest value. The interface of system development for intra-company data collection and external track and trace should also sharpen following the rapid evolution of tracking tools. An ambitious effort starting with and collaboration among tier 1 and 2 suppliers and end-of-life actors would be a strong foundation on which to expand traceability to higher tier suppliers. As such, a global data exchange protocol may provide much-needed direction. The use of existing works, such as the UN Transparency Protocol, offers prototypes to further build upon.

4. Developing incentives schemes

Beyond regulatory drivers, financial mechanisms, such as innovative incentive model and competitive procurement schemes, are crucial to supporting primary data collection and encouraging the development of collaborative IT infrastructure for data exchange with value chain participants. For example, closed-loop systems can create value through process improvements that reduce operational costs and enhance supply chain efficiency. The product-as-a-service business model could enable businesses to retain ownership of products and accelerate the adoption of second-hand, reuse and remanufacturing models, while generating additional revenue through leasing or subscription-based services.

5.2 Recommendations on immediate next steps for CDX

Building on these conclusions, we recommend WBCSD's CDX workstream focus on the following steps to enable the exchange of consistent, comparable and credible circularity data.

- **Crystalizing circularity data specification** and **circularity measurement**
  - Indicate and utilize data sought after by regulation as groundwork in defining circularity data to facilitate adoption and prevent double work.
  - Leverage the Global Circularity Protocol to harmonize industry-agnostic circularity standards: facilitate cross-sector dialogue and build on leading, established standards and definitions to unify definitions of circularity data, data categories and circularity performance measurement approaches/methodologies.
  - Define industry- or sector-specific standards adjacent to industry-agnostic circularity standards to facilitate cross-sector dialogue and collaboration with prominent industry leaders.

- **Building technology enablers**
  - Stay informed about evolving traceability systems to identify the technical solutions to facilitate intra-company and external track-and-trace capabilities.
  - Align with industry-specific databases to enhance data accuracy and encourage open access for greater adoption and improved data accuracy.
  - Collaborate with leading global data exchange initiatives and align with frameworks, protocols and data models to ensure interoperability.

- **Fostering circular business models for tech adoption**
  - Collaborate with entities already implementing platform or e-commerce-based models and existing circular schemes (such as takeback programs, in-house/external repair, refurbish models) to identify data and technology requirements to enable such models effectively.
  - Explore and use impact metrics to evaluate the benefits of alternative or circular business models, providing evidence to justify investments and encourage their adoption.

Appendices

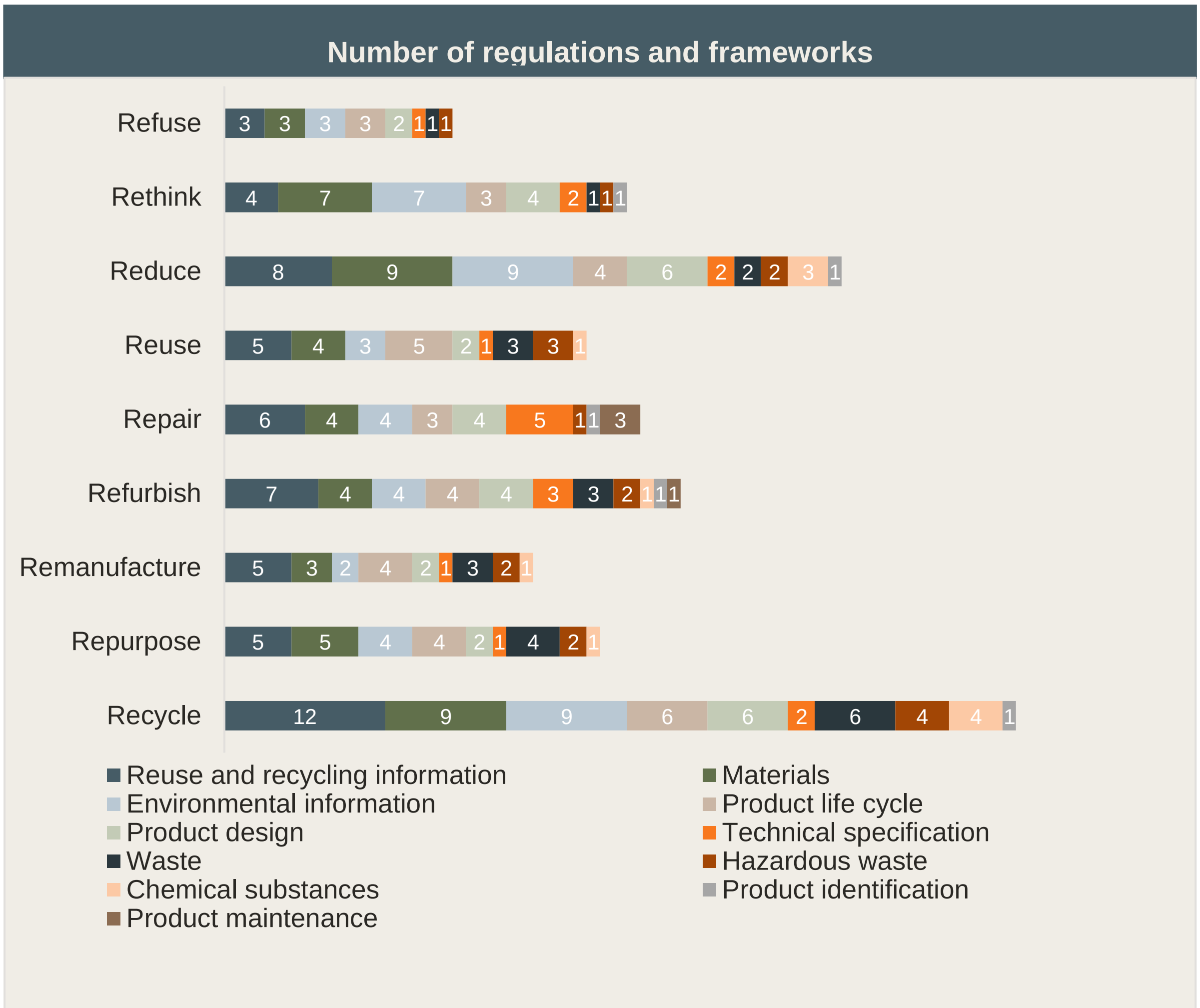
1: Regulations and voluntary frameworks considered in the landscape analysis.

No.	Name	Document type
1	California, New York, Minnesota Right to Repair state laws	Regulation
2	Carbon Broader Adjustment Mechanism	Regulation
3	Cradle to Cradle	Framework
4	Circular Transition Indicators (CTI)	Framework
5	Eco design for sustainable products regulation (ESPR)	Regulation
6	EN 15343:2008 European standard for plastics recycling traceability and recycled content	Framework
7	End-of-life vehicles directive	Regulation
8	EU Battery Regulation	Regulation
9	EU Critical Raw Materials Act	Regulation
10	EU Digital Product Passport (Part of ESPR)	Regulation
11	EU Directive on common rules promoting the repair of goods amending (EU) 2017/2394, Directives (EU) 2019/771 and (EU) 2020/1828	Regulation
12	EU Right to Repair Directive	Regulation
13	EU Single-Use Plastics Directive	Regulation
14	EU Waste Framework Directive	Regulation
15	EV battery carbon footprint disclosure requirement Japan	Regulation
16	International Sustainability & Carbon Certification (ISCC) EU certification	Framework
17	International Organization for Standardization (ISO) 14025, 14027, 59004, 59010, 59020, 59040	Framework
18	Packaging and Waste Directive	Regulation
19	Product Environmental Footprint (PEF)	Framework
20	Roundtable on Sustainable Biomaterials Association (RSB) EU Renewable Energy Directive (RED) certification	Framework
21	Sustainable Green Claims	Regulation
22	Waste Electrical and Electronic Equipment (WEEE) Directive	Regulation
23	Waste Framework Directive Update	Regulation

2: Discrepancies in information requirements of what constitutes product circularity data

Battery Passport	Product Circularity Data Sheet (PCDS)	Digital Product Passport (DPP)
General information: - Manufacturing info - Battery category - Battery weight - Battery status	**Product and company information:** - Product identifiers - Manufacturing identification - Product site information - PCSD issuance, revision	**General information:** - Unique product identifier - Global trade identification number - Relevant commodity codes - Compliance documentation
Labels and certifications - Symbols and labels - Meaning of labels - Declaration of conformity - Compliance of test results	**Composition/information on products constituents** - Chemical substance threshold - Product composition disclosure - Chemical composition - Hazard statements - Pre-consumer recycled content - Post-consumer recycled content - Sourcing statements	**Composition of product** - Requirements related to substances of concern
Carbon footprint - Carbon footprint - Weblink to CF study - CF performance class	**Design for better use** - Designed for maintenance and repair - Designed for safe operation - Designed for actively positive impacts	**Information on the performance of the product** - Repairability score - Durability score - Carbon footprint - Environmental footprint
Supply chain due diligence - Due diligence report	**Design for disassembly** - Demounting - Disassembling - Dismantling	**Information for customers and other actors on how to install, use, maintain or repair the product**
Materials and composition - Hazardous substances - Battery chemistry - Critical raw materials - Material used in cathode, anode, electrolyte	**Design for re-use** - Circularity pathways/scenarios	**Information for treatment facilities on disassembly, reuse, refurbishment, recycling, or disposal at end-of-life**
Circularity & resource efficiency - Recycled content shares - Manuals for removal, disassemble, dismantling - Component part numbers & share parts information - Safety measures, instructions		**Other information that could influence sustainable product choices for customers**
Performance & durability - Capacity, energy, power, SoH - Expected lifetime - Negative events		

3: The number of regulations requesting information under the identified data categories, mapped along the 9 R-strategies



4: Overview of the data initiatives included in the analysis

No.	Data Initiative	Purpose of the Data Initiatives
1	Catena X data ecosystem	A platform that helps organizations securely and efficiently exchange data with partners, customers or other stakeholders
2	GS1 Global Data Synchronisation Network (GDSN)	Ensures trading partners have immediate access to the most current and complete product information needed for local and global markets
3	Japan's Ouranos	Works on architecture design, R&D, demonstration, social implementation and dissemination of a system for linking multiple information
4	Gaia-X	Establishes a secure and transparent data infrastructure platform for business, promoting data sovereignty and digital autonomy
5	Global Data Service Organization for tires and automotive components (GDSO)	Standardizes data related to tires, defines solutions to access and exchange data and develops online solutions
6	GS1 Global Data Model	Standardizes data exchange in supply chains and retail sectors globally, improving efficiency and accuracy in information sharing
7	Luxembourg's Product Circularity Data Sheet (PCDS)	Facilitates data exchange and collaboration in financial sector of Luxembourg, enhancing transparency and efficiency in financial services
8	Union database (UDB)	From 2024, all trade of biofuels and biomass subject to the Renewable Energy Directive (RED III) regulation will have to submit information to the EU-managed union database
9	United Nations Transparency Protocol (UNTP)	Promotes transparency in sustainability and environmental aspects according to UN guidelines, facilitating accountability and reporting practices globally
10	ECHA (European Chemicals Agency) CHEM	A central database where manufacturers and importers must register information on the properties of their chemical substances
11	International Material Data System (IMDS)	Collects, maintains, analyses and archives materials used for automobile manufacturing
12	China Automotive Material Data System Software (CAMDS)	The Chinese equivalent of IMDS, focusing on material data for automobile manufacturing
13	European Product Registry for Energy Labelling (EPREL)	Facilitates the registration of products requiring an energy label before their placement on the EU market
14	Boost 4.0	A European initiative in big data for industry 4.0, aligning with the International Data Spaces (IDS) reference architecture model to leverage private investment
15	Raw Materials Information System (RMIS)	Provides knowledge on raw materials and associated value chains, considering sustainability, supply risks and circularity
16	Ecoinvent	Maintains a comprehensive life-cycle inventory database that provides reliable and transparent information on the environmental impact of various products
17	Green Deal Dataspace	Enables secure and efficient sharing of environmental sustainability data to foster green technologies and resource efficiency

5: Glossary

Circular accounting: The practice of measuring and analyzing a company's financial and non-financial performance to truly reflect and account for the value and impact of circular businesses (adapted from the Coalition Circular Accounting[20]).

Product circularity data: The standardized information on the circularity aspects of a product that other stakeholders could use partially or entirely to enable circular evaluation of the product (adapted from ISO PCDS 59040[21]). For this reason, product circularity data can be information collected and/or measured.

Circularity data: The structured information that captures key metrics related to an organization's or product's circular performance (e.g., reduce waste and resource consumption, maximizing closed loops).

9 R-strategies: Classification of circular strategies aimed at reducing the amount of materials and resources consumed for product production, by focusing on product use and manufacture (R1-R2); material life extension (R3 – R8), and end-of-life recovery (R9).

(R1) Refuse: Make solutions redundant by abandoning its function or by offering the same function with a radically different solution (ISO 59004[22]).

(R2) Rethink: Reconsider design and manufacturing decisions. Make service use more intensive (e.g. through sharing or by putting multi-functional products on the market) (ISO 59004[23]).

R3) Reduce: Increase efficiency in product manufacture or use by consuming fewer natural resources and materials (ISO 59004[24]).

(R4) Reuse: To extend a product's lifetime beyond its intentional designed life span, without changes made to the product or its functionality (CTI[25]).

(R5) Repair: To extend a product's lifetime by restoring it after breakage or tearing, without changes made to the product or its functionality (CTI[26]).

(R6) Refurbish: To extend a product's lifetime by large repair, potentially with replacement of parts, without changes made to the product's functionality (CTI[27]).

(R7) Remanufacture: To disassemble a product to the component level and reassemble (replacing components where necessary) to as-new condition with possible changes made to the functionality of the product (CTI[28]).

(R8) Repurpose: Adapt a product or its component parts for use in a different function than originally intended without making major modifications to its physical or chemical structure (ISO 59004[29]).

(R9) Recycle: To reduce a product back to its material level, thereby allowing the use of those materials in new products (CTI[30]).

6: Acronyms

API	application programming interface
CDX	Circularity Data Exchange
CTI	Circular Transition Indicators
CODES	Coalition for Digital Environment Sustainability
CSRD	Corporate Sustainability Reporting Directive
DPP	Digital Product Passport
D4CE	Digitalization 4 Circular Economy
ESPR	Ecodesign for Sustainable Products Regulations
EOL	end of life
EU	European Union
EPR	extended producer responsibility
GBA	Global Battery Alliance
GCP	Global Circularity Protocol for Business
GHG	greenhouse gas
GHGP	Greenhouse Gas Protocol
IP	intellectual property
IMDS	International Material Data System
MCI	Material Circularity Indicator
OPN	One Planet Network
OEM	original equipment manufacturer
PACT	Partnership for Carbon Transparency
PCF	Product Carbon Footprint
PEF	Product Environmental Footprint
UNEP	United Nations Environment Programme
WEEE	Waste Electrical and Electronic Equipment Directive

Endnotes

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Acknowledgments

Disclaimer

This guide is released in the name of WBCSD. The Circularity Data Exchange (CDX) members and other experts reviewed draft, ensuring that these recommendations broadly represent the majority of project members' views. It does not mean, however, that every CDX member and all expert organizations agree with every word.

Supplementary information to this White Paper can be made available on a demand basis.

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WBCSD is a global community of over 220 of the world's leading businesses, representing a combined revenue of more than USD $8.5 trillion and 19 million employees. Together, we transform the systems we work in to limit the impact of the climate crisis, restore nature and tackle inequality.

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